

second step was to select the cited references in the publications from 1980 to 2012 and to determine the number of cited references per year (from 1650 to 2012) (755,607,107 cited references in total). The annual number of publications or cited references formed the basis for the (segmented) regression analyses (van Raan, 2000) – the third step in the analysis.

Based on the annual number of publication, a growth model  $y(t)=b_0*\exp(b_1*(t-1980))$  was estimated by a nonlinear regression using SAS PROC NLIN (SAS Institute Inc., 2011), where the intercept  $b_0$  equals  $y(0)$  – the outcome in the year 1980. The model converged: overall 96% of the total variance of the annual number of publications could be explained by the regression model.

Segmented regression analysis was used to determine different segments of growth development in the cited references within the annual time series (Bornmann, Mutz, & Daniel, 2010; Brusilovskiy, 2004; Lerman, 1980; McGee & Carleton, 1970; Mutz, Guilley, Sauter, & Nepveu, 2004; Sauter, Mutz, & Munro, 1999; Shuai, Zhou, & Yost, 2003). In the model estimations, the logarithmised number of cited references per year forms the dependent variable. In mathematical and statistical terms we assumed a simple exponential growth model which considers separate segments in the time series (e.g., a segment with a decline around both World Wars, WW). This model can be formulated as a differential equation  $f'(t)=b_1*f(t)$  where  $b_1$  is the growth constant or the multiplication factor and  $t$  is the time (cited reference year). The change  $f(t)$  in a period  $t_1-t_0$  is therefore proportional to the status at the starting point in time  $t_0$ . The solution of the differential equation is an exponential function:  $y(t)=y(0)*\exp(b_1*t)$ . The growth rate in percent  $(y(t)-y(0))/y(0)$  is  $\exp(b_1)-1$ . The doubling time is the amount of time required for an outcome to double in size ( $t=\ln(2)/b_1$ ). Logarithmising the function  $y(t)$  results in a linear function  $\ln(y)=b_0+b_1*t$  with  $b_0=\log(y(0))$ , the parameters of which can be estimated with a linear regression.

In the segmented regression different segments can be identified with different regression coefficients, where both the breakpoints (cited reference year) and the segments as

well as the growth constant  $b_1$  of each segment is estimated. For example, let  $\log\_y$  the logarithmic transformed annual number of cited references, ‘year’ the cited reference year, and  $a_1/ a_2$  the breakpoints of differentiating three segments. Then, we need to estimate the unknown regression parameters  $b_0, b_1, b_2$  and  $b_3$ , and the breakpoints  $a_1$  and  $a_2$  by minimizing the following objective function – in particular the sum of squared residuals (Brusilovskiy, 2004, p. 2):

$$\begin{aligned}
 &\text{IF year} \leq a_1 \text{ THEN } \log\_y = b_0 + b_1 * \text{year} + \varepsilon; \\
 &\text{ELSE IF year} \leq a_2 \text{ THEN } \log\_y = b_0 + b_1 * a_1 + b_2 * (\text{year} - a_1) + \varepsilon; \\
 &\text{ELSE IF year} \leq a_3 \text{ THEN } \log\_y = b_0 + b_1 * a_1 + b_2 * (a_2 - a_1) + b_3 * (\text{year} - a_2) + \varepsilon \\
 &\varepsilon \sim \text{i.i.d. } N(\mathbf{0}, \mathbf{I}\sigma^2)
 \end{aligned} \tag{1}$$

In the function the residuals  $\varepsilon$  are assumed to be multivariate normally distributed with a zero mean vector and a covariance matrix with identical variances  $\sigma^2$  (homoscedasticity) and zero covariance (no autocorrelations of the residuals) overall and across the segments. These are rather strong assumptions for time series data. Given the high proportion of explained variance (low proportion of residual variance in this study) and low standard errors, which we generally observed in the regression analyses, the violations of the assumptions were not considered with the risk of slightly higher standard errors. The model parameters were estimated by the least squares method (Gauss-Newton) under the restriction that the breakpoints  $a_1, a_2, \dots$  are ordered (e.g.,  $1690 < a_1 < a_2 < 2012$ ). In order to avoid local minima of the estimation procedure a grid of different starting values for the parameters was used. The regression constant  $b_0$  was erased in our analysis to enhance the fit of the model.

The following criteria were used to choose the number of segments: First, by visually inspecting the data it was obvious that the relationships are strongly linear with different slopes within three or four segments (see, e.g., Figure 2). Second, statistical criteria were

used. In the case of least-squares estimations the amount of explained variance ( $R^2$ ) is the method of choice. In this study, models were selected, where the amount of explained variance no longer substantially increased. In other words, segmented regression models with three breakpoints could explain 99% of the total variance of the annual number of references. Thus, it was possible to divide the series of the annual number of cited references in the time interval from 1650 to 2012 into four segments.

Differences between disciplines (natural sciences and medical and health sciences) (D) regarding the slope parameters are tested by expanding equation 1 by corresponding interaction terms with disciplines (e.g.,  $b_1 \cdot D \cdot \text{year}$ ) for each slope parameter ( $b_1, b_2, b_3, \dots$ ). The disciplines were dummy-coded (natural sciences:  $D=0$ , medical and health sciences:  $D=1$ ).

The statistical analyses were performed using the SAS procedure PROC NLIN (SAS Institute Inc., 2011).

### 3 Results

Figure 1 shows the exponential growth of global scientific publication output for the period 1980 to 2012 (33 years). While the dots represent the observed values, the predicted values are the result of the regression analysis. As the result of the segmented regression analysis (Table 1) shows, the global scientific publication output is growing at a rate of approximately 3% annually. The volume of publications doubles approximately every 24 years. A similar result – exponential growth of publications – is also reported by Pautasso (2012) for 18 biological sub-fields.

However with the model in Table 1 we are looking at a relatively short and recent period of scientific activity in society. The analysis of cited references from the publications allows us to start the analysis of scientific activity much earlier. According to van Raan (2000) scientific evolution started “after the Scientific Revolution in 16th century Europe” (p.

347). However, van Raan (2000) starts the analysis of cited references considerably later, in 1800. In this study, we start with the analysis of cited references in the middle of the 17th century. This period saw the development of the institutionalised structures of modern science with publication of the results of scientific work in journals and manuscripts undergoing critical peer review before publication (Bornmann, 2011). In peer review, active scientists in the same discipline review the scientific work of each other and decide on success and failure. The use of the peer review procedure since the middle of the 17th century is an important step towards an independent science system from other areas of society, for example from church (Gould, 2013). According to Popper (1988) we can assume that it is rational criticism – institutionalized as peer review procedures – driven by the idea of truth that characterises modern science. Gattei (2009) says that the "growth of knowledge and criticism are closely interconnected" (p. 2).

Figure 2 shows the segmented growth of the annual number of cited references between 1650 and 2012. The observed values and those predicted by the model are visualised. As Table 2 shows, regression analysis has identified four different segments with different growth rates: The first phase from 1650 to approximately 1750 is characterised by a relatively low growth rate of less than 1%, with cited references doubling in 150 years approximately. It is followed by a phase with a growth rate of slightly less than 3% which lasts until the period between WWI and WWII. Interestingly, these two segments identified by the regression analysis correspond to phases of economic growth and the start of the Industrial Revolution in Europe in about 1750. "Pre-1750 growth was primarily based on Smithian and Northian effects: gains from trade and more efficient allocations due to institutional changes. The Industrial Revolution, then, can be regarded not as the beginnings of growth altogether but as the time at which technology began to assume an ever-increasing weight in the generation of growth and when economic growth accelerated dramatically. An average growth rate of 0.15–